

SMART INDIA HACKATHON 2026

MINEGUARD AI

Smart Mine Safety and Autonomous HEMM Fleet Coordination System

Simulation-Based Prototype for Safe and Efficient Mine Vehicle Operation

Problem Statement: SIH26007

Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast
Iron Ore Mines

Ministry of Steel • NMDC • Smart Automation • Hardware

ROS 2 Jazzy • Gazebo Sim 8.11.0 • Blender 5.2.1 • Python • OpenCV

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1. Executive Summary

This project presents a simulation-based smart mine safety and autonomous haulage system for Heavy Earth Moving Machinery (HEMM) operating in an open-cast iron ore mine under severe fog and low-visibility conditions. A digital open-pit mine and a fleet of three autonomous HEMM dump trucks were developed using ROS 2 Jazzy and Gazebo Sim 8.11.0.

Each simulated HEMM is equipped with LiDAR, GPS, IMU, RGB camera and depth sensing. The system combines sensor-based safety assessment, waypoint-based autonomous route following, operator target-speed control, deterministic Vehicle-to-Vehicle (V2V) coordination and a centralized real-time control-room dashboard.

The safety layer continuously evaluates obstacle clearance and assigns graduated risk states from CLEAR to CRITICAL. It can reduce the commanded speed or stop a vehicle when a critical hazard is detected. The V2V layer exchanges fleet state through ROS 2 and applies deterministic right-of-way rules at shared conflict zones. A non-priority vehicle can wait with zero motion until the priority vehicle clears the conflict zone.

The prototype is intentionally simulation-first. It validates the software architecture, safety logic, coordinated fleet behavior and monitoring workflow before future integration with real mining vehicles and industrial-grade sensing and communication equipment.

2. Official SIH Problem Statement

Problem Statement Number: SIH26007

Official Title: Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines

The problem addresses the safety and productivity challenges faced by HEMM dumpers in the Bailadila mining region during severe monsoon conditions. Dense fog can reduce visibility on haul roads to approximately 3–5 metres, increasing the risk of collisions and operational disruption. Reduced movement can also increase haul cycle time, reduce fleet productivity and affect ore evacuation.

The intended solution direction calls for improved operator situational awareness, vehicle guidance and collision avoidance, real-time monitoring and decision support during adverse weather. Relevant technologies identified by the problem include AI/ML analytics, computer vision and thermal imaging, LiDAR/radar sensing, GPS/DGPS, V2V/V2I communication, autonomous or driver-assistance systems, IoT monitoring, centralized command and control and digital-twin monitoring.

Expected outcomes include improved dumper safety, reduced collision risk, better haulage efficiency, improved fleet utilization and continuity, reduced losses caused by low visibility, real-time monitoring and decision support, and a scalable solution for large mechanized open-cast mines.

3. Project Objectives

- Create an open-pit mine simulation with benches, haul roads, ramps, loading and dumping areas, turning areas and physical hazards.
- Simulate three autonomous HEMM dump trucks as a coordinated fleet.
- Provide multi-sensor situational awareness using LiDAR, GPS, IMU, RGB camera and depth camera data.
- Detect obstacles and classify vehicle safety into CLEAR, ADVISORY, CAUTION, WARNING and CRITICAL states.
- Implement autonomous waypoint-based route following with operator target-speed control.
- Implement deterministic V2V right-of-way coordination at predefined conflict zones.
- Reduce skidding risk by preventing large steering corrections while the vehicle is moving rapidly.
- Provide a real-time control-room dashboard for position, speed, safety, V2V state, alerts and camera monitoring.
- Keep the safety layer independent from normal navigation so that safety can override movement commands.

4. Proposed Solution

The proposed solution is a digital mine and autonomous fleet prototype integrated through ROS 2.

1. Gazebo provides the mine environment, roads, ramps, benches, loading/dumping zones and hazards.
2. Each HEMM publishes simulated sensor and vehicle-state information through ROS 2.
3. The safety system processes LiDAR information and determines obstacle clearance and risk level.
4. The autonomous controller follows predefined mining routes using waypoints and heading control.
5. The operator supplies a target cruising speed; the controller and safety governor determine the final safe command.
6. The V2V layer exchanges vehicle state and coordinates right-of-way at conflict zones.
7. The dashboard receives live ROS 2 telemetry and presents fleet status, map positions, cameras, alerts and V2V states.

Control hierarchy:

Operator target speed → Autonomous route controller → V2V traffic permission → Safety governor → Final velocity command.

The safety governor has the highest priority. A critical safety condition can stop the vehicle even if normal route following or V2V logic would otherwise permit movement.

5. System Architecture

The system is organized into modular layers:

- Mine Simulation: open-pit terrain, haul roads, ramps, benches, loading/dumping areas and hazards.
- HEMM Simulation: three separately controlled autonomous dump trucks.
- Sensing: LiDAR, GPS, IMU, RGB camera and depth camera.
- Safety Layer: obstacle-distance evaluation, risk classification and speed/stop intervention.
- Autonomous Navigation: waypoint following, heading control and target-speed management.
- V2V Coordination: vehicle-state exchange, conflict detection, right-of-way, waiting and release.
- ROS 2 Communication: topic-based communication between sensors, controllers, safety, V2V and dashboard.
- Dashboard: live fleet map, telemetry, cameras, alerts and V2V monitoring.

6. Tools and Technologies Used

ROS2 Jazzy — robot middleware, nodes, topics, sensor/control communication and fleet coordination.

Gazebo Sim 8.11.0 — physics-based mine and HEMM simulation, sensor simulation and testing.

Blender 5.2.1 — open-pit mine environment and visual assets.

Python — autonomous controller, safety system, V2V logic, dashboard server and utilities.

ros_gz_bridge — ROS 2 ↔ Gazebo message bridging.

OpenCV — image processing for the simulated thermal-style visualization.

HTML/CSS/JavaScript — control-room dashboard interface and live map visualization.

Ubuntu 24.04.4 LTS — development and execution environment.

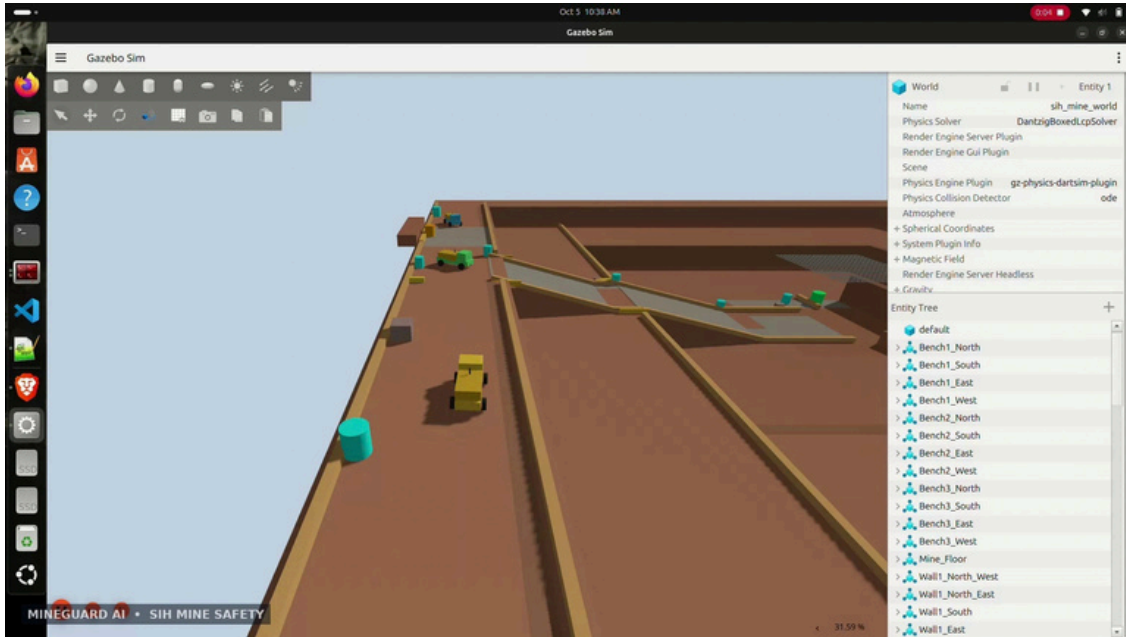


Figure 1. MINEGUARD AI simulation and mine-fleet monitoring view.



Figure 2. MINEGUARD AI simulation and mine-fleet monitoring view.

7. Sensors and Interfaces

LiDAR: 360° obstacle-distance sensing for situational awareness and safety assessment.

GPS: simulated global vehicle position.

IMU: orientation and motion information.

RGB Camera: forward visual monitoring and dashboard camera feed.

Depth Camera: simulated depth information from the environment.

V2V State Interface: exchange of vehicle position, speed, heading, route and traffic state.

Important implementation note: the dashboard thermal view is a simulated thermal-style visualization generated from RGB data. It is not claimed to represent calibrated physical FLIR measurements.

8. Major Modules Developed

8.1 Mine Environment

The environment contains multiple bench levels, haul roads, connecting ramps, loading and dumping areas, turning areas, vehicle spawn regions and physical hazards such as rocks, barriers and berms.

8.2 HEMM Fleet

Three separately namespaced vehicles — HEMM-01, HEMM-02 and HEMM-03 — were implemented with independent command, odometry, GPS, IMU, LiDAR and camera topics. An operator target-speed interface is provided, while the safety layer constrains the final velocity.

8.3 Safety and Collision-Risk System

LiDAR front/corridor measurements are continuously evaluated. Risk states progress from CLEAR through ADVISORY, CAUTION and WARNING to CRITICAL. Increasing risk progressively constrains speed, while CRITICAL conditions can command a complete stop.

8.4 Autonomous Navigation

Lightweight waypoint-based navigation was implemented for the mine circuit. Heading error controls steering, moderate heading errors reduce speed, and large heading errors cause forward motion to stop while the vehicle aligns.

8.5 V2V Fleet Coordination

The current V2V layer is deterministic and rule-based rather than an ML model. Vehicles exchange state through ROS 2 fleet topics. At a conflict zone, one vehicle receives right-of-way and another enters a waiting state with zero linear and angular velocity. After the priority vehicle clears the zone, the waiting vehicle is released. Safety remains higher priority than V2V permission.

8.6 Control-Room Dashboard

The dashboard provides a real-time three-vehicle overview, 2D mine map, vehicle selection, target-speed controls, RGB and simulated thermal views, LiDAR clearance, safety state, V2V coordination state, event logs and navigation telemetry. Values are sourced from the ROS 2 simulation rather than manually entered.

9. ROS 2 Communication Structure

Representative topics include:

/hemm_01/cmd_vel, /hemm_02/cmd_vel, /hemm_03/cmd_vel

/hemm_01/odom, /hemm_02/odom, /hemm_03/odom

/hemm_01/scan, /hemm_02/scan, /hemm_03/scan

/hemm_01/gps, /hemm_02/gps, /hemm_03/gps

/hemm_01/imu, /hemm_02/imu, /hemm_03/imu

/hemm_01/camera/image_raw and corresponding camera/depth topics

/hemm_01/safety/risk_level

/hemm_01/safety/warning

/hemm_01/safety/obstacle_distance

/fleet/vehicle_states

Equivalent namespaced topics are used for HEMM-02 and HEMM-03. The fleet topic provides the state needed for coordinated traffic decisions and dashboard synchronization.

10. Testing and Validation

The prototype was tested in the following areas:

- Gazebo world launch and mine loading.
- HEMM physical stability and movement.
- Forward, reverse and turning behavior.
- Ramp movement and route transitions.
- LiDAR scan and front/corridor obstacle measurements.
- Graduated safety levels from CLEAR to CRITICAL.
- Critical safety stop behavior.
- Fleet-state and V2V topic communication.
- Right-of-way, crossing, waiting and release states.
- Anti-skid behavior for large heading corrections.
- Live dashboard telemetry and camera streams.
- Live vehicle-marker rendering and coordinate synchronization.

The recorded demonstration shows the three HEMMs moving through the simulated mine while the dashboard updates fleet position, speed, heading, safety state, V2V coordination and camera information in real time.

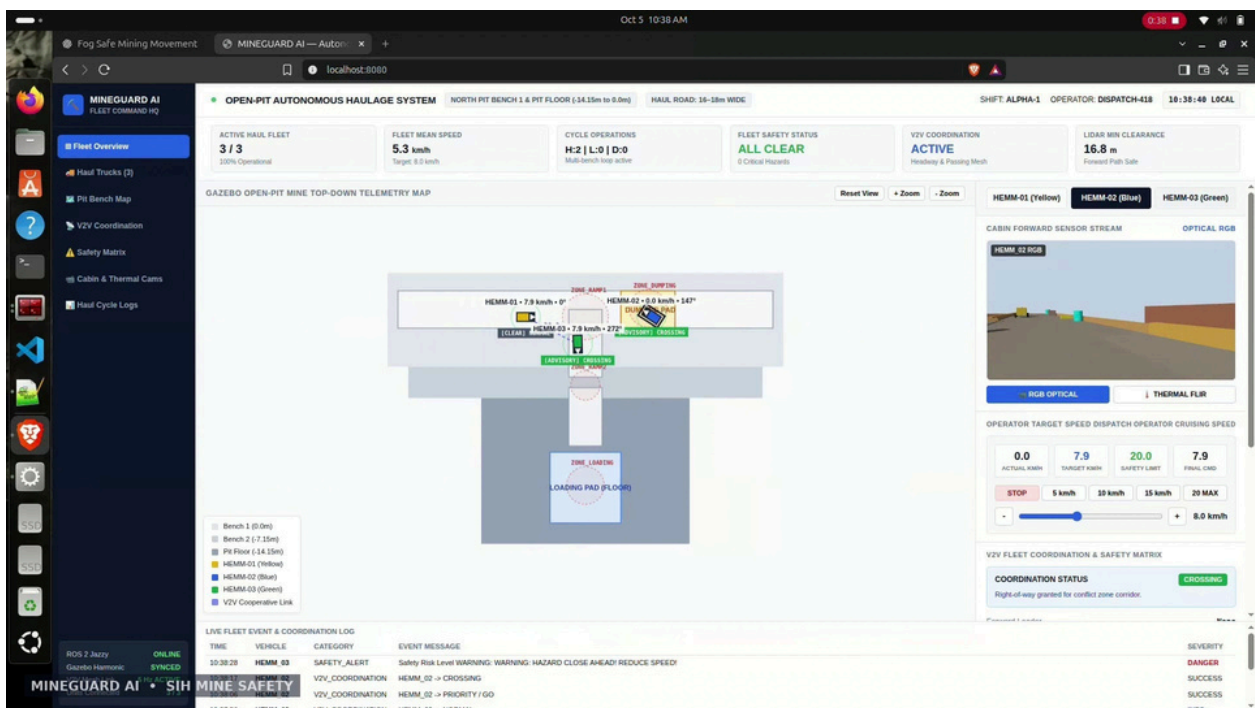


Figure 3. Real-time fleet dashboard with V2V and safety telemetry.

11. Key Prototype Outcomes

- A complete simulation environment for HEMM safety and coordinated haulage was developed.
- Three autonomous HEMM vehicles can be represented and monitored as a fleet.
- Multi-sensor simulated situational awareness is available through LiDAR, GPS, IMU, RGB and depth sensing.

- A safety governor provides graduated risk states and can reduce or stop vehicle motion.
- Autonomous waypoint-based route following was implemented for the mine circuit.
- Anti-skid control logic was added so that large heading corrections do not occur simultaneously with high forward speed.
- Deterministic V2V right-of-way coordination was implemented for conflict zones.
- A non-priority vehicle can be placed into a complete waiting state until the priority vehicle clears the zone.
- A control-room dashboard provides live fleet telemetry, map visualization, camera monitoring, safety status and V2V state.
- The architecture provides a practical simulation baseline for future hardware and field integration.

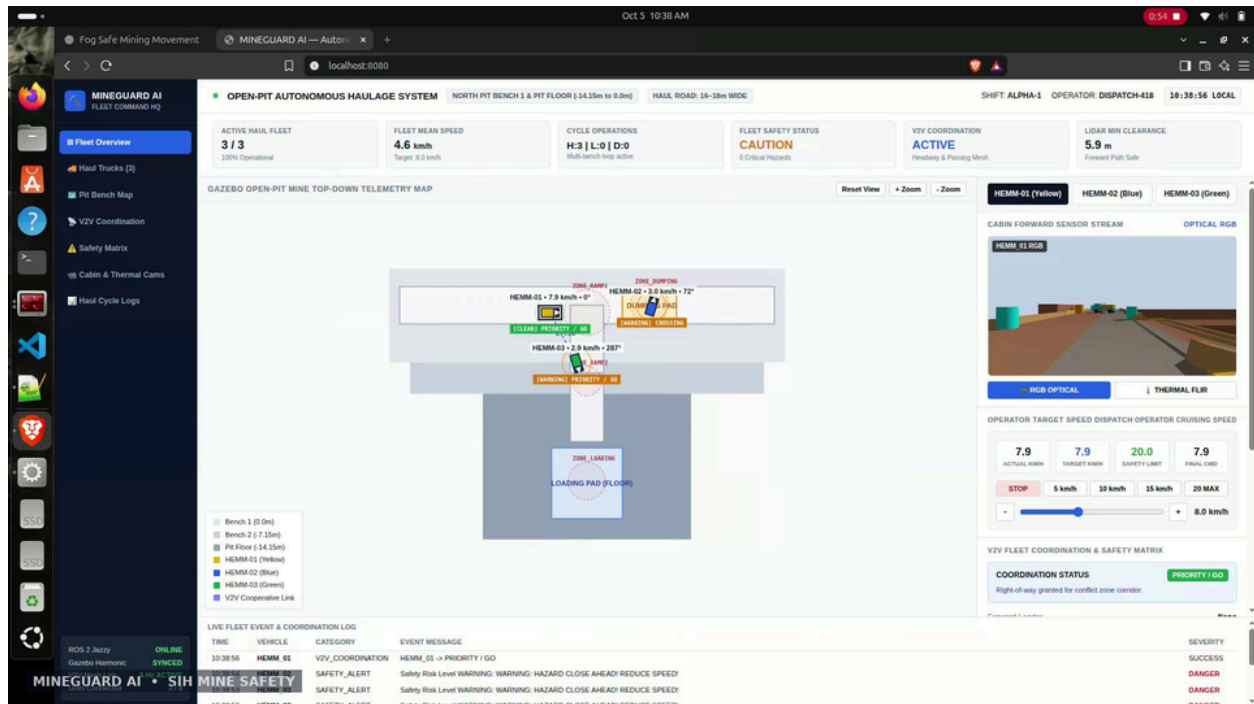


Figure 4. Live simulation monitoring view.

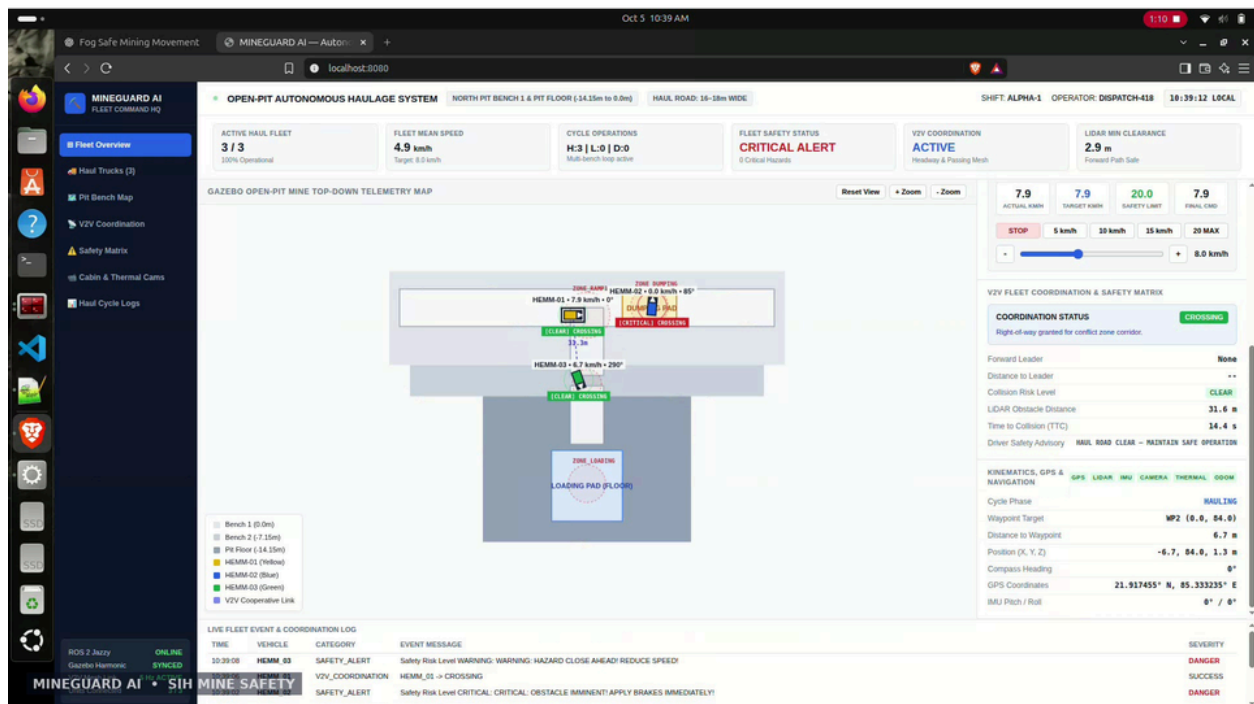


Figure 5. Live simulation monitoring view.

12. Alignment with SIH Expected Outcomes

Improved dumper safety — LiDAR-based safety monitoring, graduated risk states and safety-stop behavior.

Reduced collision risk — obstacle detection combined with V2V right-of-way at conflict zones.

Improved haulage efficiency — autonomous route following, target-speed control and coordinated movement.

Fleet utilization and continuity — three-vehicle fleet simulation and traffic coordination.

Reduced low-visibility operational losses — sensor-based situational awareness and controlled movement in simulated adverse conditions.

Real-time monitoring and decision support — centralized dashboard with live map, telemetry, cameras, safety and V2V information.

Scalability — ROS 2 namespacing and modular vehicle nodes allow additional vehicles and real sensors to be integrated later.

13. Technical Innovation

- Simulation-first digital mine approach for safe and repeatable development.
- Multi-HEMM coordination rather than a single-vehicle demonstration.
- Deterministic V2V right-of-way logic for predictable traffic behavior.
- Independent safety governor with priority over normal navigation and traffic permission.
- Anti-skid steering behavior that separates vehicle alignment from high-speed forward motion.
- Live mine-map visualization synchronized with fleet telemetry.
- Integration of autonomous navigation, local safety sensing and fleet-level coordination in one prototype.

14. Current Limitations

- The current implementation is a simulation prototype and has not been field-validated on a real HEMM.
- Radar sensing is not part of the currently validated sensor stack.
- The thermal stream is a simulated visualization rather than calibrated thermal-camera data.
- Autonomous navigation uses lightweight waypoint following rather than a full industrial mine-navigation stack.
- V2V is currently implemented through ROS 2 simulation communication; real deployment would require industrial wireless/V2X communication and cybersecurity controls.
- Real DGPS/RTK positioning, vehicle interfaces, functional-safety certification and mine-specific operational constraints would need to be addressed before deployment.

15. Future Scope

- Integrate industrial-grade LiDAR and long-range radar for adverse-weather sensing.
- Integrate calibrated thermal cameras for night and low-visibility operation.
- Use RTK/DGPS for higher-precision positioning and map alignment.
- Develop real V2V/V2I communication using suitable industrial wireless or V2X technology.
- Add AI/ML models for object detection, classification, risk prediction and predictive maintenance where appropriate.
- Integrate the prototype with a mine control center and fleet-management platform.
- Develop a high-fidelity digital twin synchronized with real mine telemetry.

- Test with hardware-in-the-loop and scaled HEMM/RC platforms before field trials.
- Add cybersecurity, fail-safe communication handling and formal safety validation for production deployment.

16. Conclusion

The developed MINEGUARD AI prototype addresses the central SIH26007 challenge by combining autonomous HEMM movement, sensor-based safety monitoring, deterministic V2V coordination and centralized real-time monitoring in one simulation environment.

The system demonstrates how a fleet of mine vehicles can maintain situational awareness, coordinate at shared conflict points and apply safety-based speed or stop decisions. The simulation-first approach provides a controlled and economical environment for validating safety logic and fleet behavior before moving toward physical implementation. The modular ROS 2 architecture provides a clear path for future integration of industrial sensors, real V2V/V2I communication, higher-precision positioning, AI-based perception and real mine-control systems.

Demonstration Video

The project submission includes MINEGUARD_AI_SIH_Demo.mp4, showing the Gazebo mine simulation, three HEMMs, live dashboard telemetry, safety states, V2V coordination and camera monitoring.

17. Viva Quick Reference

What is the project? A smart mine safety and autonomous HEMM fleet coordination system for low-visibility conditions.

PS number? SIH26007.

Organization? Ministry of Steel / NMDC.

Core platform? ROS 2 Jazzy + Gazebo Sim 8.11.0.

How many vehicles? Three autonomous HEMM dump trucks.

Main sensors? LiDAR, GPS, IMU, RGB camera and depth camera.

How does V2V work? Deterministic ROS 2 state exchange and conflict-zone right-of-way coordination.

Is V2V AI/ML? No. The current V2V decision layer is deterministic and rule-based.

Who has highest priority? The safety layer.

What happens at a crossing? One HEMM gets right-of-way; the other stops and resumes after the zone is clear.

Why simulation? To validate architecture and safety behavior safely and economically before hardware deployment.

Is thermal real? No. It is a simulated thermal-style visualization from RGB data.

Future AI use? Object detection, risk prediction and advanced perception/prediction.

18. Reference

Smart IndiaHackathon2026 Problem Statement SIH26007, Ministry of Steel / NMDC: "Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines."

Project implementation details are documented from the developed ROS 2, Gazebo, Blender, Python and dashboard prototype.